Ari Gestetner

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Education

Monash University | ***Melbourne, Victoria, Australia*** **2023-2025**
Bachelor Science in Mathematics and Computational Science

- Awarded with summer and winter vacation research scholarships
- First place in an IMC algorithmic trading competition for Monash students hosted by Monash Association of Coding (MAC)

Experience

Full-Stack Developer **May 2026 - Present**
Bolder Solutions *New York City, NY* *Vue, Laravel, MySQL*

- Full-stack development on a camp and school management CRM, building and maintaining features across the Vue frontend, Laravel backend, and MySQL database
- Developing and enhancing full-stack features including filterable list and edit pages, bulk communication workflows, and API integrations

Teaching Associate (Intro to Programming, Intro to Computer Science & Theory of Computation) **Feb 2024 - Present**
Monash University

- Ran tutorials, consults and provided support in lectures and workshops of up to 200 students
- Adapted explanations to each student's level, leading with intuition for abstract topics and pushing students to think through problems themselves
- Scored 91% on the official unit survey, above the 86% faculty average, and ran 250+ one-on-one assignment interviews

Research Scholarship — Dr. Buser Say **Jul 2023 - Feb 2025**
Monash University *Python, SCIP, Cplex, Gurobi, Seaborn, MILP, Constraint Programming*

- Benchmarked different solvers and LP constraint formulations for neural-network transition functions, achieving over 2x reductions in solve time
- Refactored and extended SCIPPlan, an automated planner for non-linear continuous-time problems, and developed SCIP-Plan+, an extension for discrete and continuous-time problems with stochastic domains
- Reformulated a previously intractable planning problem, cutting solve time from hours-long timeouts to under 30 minutes
- Published the work as a peer-reviewed paper, A Metric Hybrid Planning Approach to Solving Pandemic Planning Problems with Simple SIR Models, at ICAART 2026
- Extended SCIPPlan to support systems of ODEs and function interpolation for more complex planning problems
- Introduced GitHub workflows for automated packaging and deployment to PyPI, enabling fast and easy development

Projects

Retail Sales Forecasting *R, Fable, Tidyverse*

- Analysed multi-decade Queensland retail turnover data to identify long term growth trends and predictable seasonal shopping patterns
- Evaluated multiple statistical models to forecast future sales, specifically measuring the impact of major market disruptions like the 2020 pandemic and the speed of subsequent recoveries

Home Server & Infrastructure *Docker, Reverse Proxies, Linux, Networking*

- Manage a dedicated home server for self-hosting applications, utilising Docker deployment and container orchestration
- Configured VPN, reverse proxies and environments to enable secure and easy access and maintain high availability for hosted containers

Technical Skills

Languages: Python, R, Typescript, C, C++17, SQL, HTML/CSS
Frameworks: Flask, FastAPI, Django, PyTest, React, Vue, Nuxt, JUCE
Developer Tools: Git, Docker, RStudio, Linux, Vercel, GCP
Libraries: BeautifulSoup, Pandas, Matplotlib, Numpy/Scipy, Pytorch, Tidyverse, Fable, Cplex, Gurobi, SCIP
Academic Concepts: Analysis, Linear Algebra, Probability, Optimisation and OR, Graph Theory, Forecasting, Statistical Modelling, Econometric Theory